# Human State Representation for Real-Time Emotion-Aware Context Generation: A Pipeline for Fusing Speech, Emotion, and Intent Evidence

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***Abstract*—*Human communication extends beyond words, as emotions and motivation play an important role in enabling natural and meaningful interactions. However, existing Speech Emotion Recognition (SER) systems primarily focus on identifying emotions using handcrafted features and deep learning models such as HuBERT and wav2vec 2.0, treating emotion recognition as an end task rather than contextual information. Similarly, most LLM-based conversational systems rely mainly on transcripts and chat history, overlooking emotional and contextual cues that influence communication. This can lead to responses that are contextually or emotionally inappropriate. Moreover, continuous speech and context processing in real-time voice interactions can increase computational requirements and token usage. To address these limitations, this work proposes a Human State Representation (HSR) framework that integrates speech emotion representations, transcript semantics, and conversational context into a structured representation. This enriched context can guide LLMs toward more context-aware and emotionally responsive responses while supporting efficient real-time voice-to-voice interaction.*

***Index Terms*—*Speech emotion recognition, self-supervised speech representations, HuBERT, human state representation, rule-based reasoning, probabilistic scoring, real-time systems, affective computing.*

## I. INTRODUCTION

Human communication conveys information through multiple complementary signals, including spoken linguistic content, acoustic characteristics, paralinguistic cues, emotional expressions, and conversational context. Conventional speech emotion recognition (SER) systems have predominantly focused on acoustic information; however, acoustic-only approaches may not fully capture the semantic and contextual information contained in an utterance. Atmaja *et al.* highlighted the importance of integrating acoustic and linguistic information for speech emotion recognition, particularly because linguistic information can be obtained from speech transcripts generated through automatic speech recognition (ASR) systems [5]. More recent work on multimodal emotion recognition similarly emphasizes that real-world conversational systems require information from multiple modalities rather than relying on a single source of evidence [6], [7].

The development of self-supervised learning has significantly improved the extraction of meaningful representations from raw speech. HuBERT introduced a self-supervised speech representation learning approach based on masked prediction of hidden units, enabling the model to learn contextualized representations from unlabeled speech [1]. WavLM further demonstrated the potential of self-supervised representations for a broader range of speech-processing tasks by incorporating speech denoising and masked prediction during pre-training [2]. These representations have subsequently been investigated for speech emotion recognition. Morais *et al.* demonstrated that self-supervised speech features can provide effective representations for categorical emotion recognition, while Pepino *et al.* showed the usefulness of wav2vec 2.0 representations for speech emotion recognition [3], [4]. These studies establish pretrained speech representations as an effective foundation for extracting emotional information from speech.

However, identifying an emotion label alone does not necessarily provide a complete representation of a person's conversational state. For example, the same emotional expression may occur in different conversational situations, while identical linguistic content can be expressed with substantially different emotional characteristics. Conversation-level emotion recognition therefore requires consideration of the speaker's previous interactions and the evolving conversational context. DialogueRNN demonstrated the importance of modeling conversational context and speaker-specific states for emotion detection in conversations [13]. Similarly, research on empathetic dialogue emphasizes that recognizing a conversation partner's feelings is an important component of effective conversational interaction [10]. Liu *et al.* further showed that effective emotional support involves not only recognizing emotional situations but also selecting appropriate conversational support strategies [11].

Automatic speech recognition provides an additional source of linguistic evidence for such systems. Whisper demonstrated that large-scale weakly supervised training can produce robust speech recognition models across diverse audio conditions and languages [8]. Consequently, ASR can provide a textual representation of an utterance that complements the acoustic information extracted directly from speech. Nevertheless, incorporating multiple speech-processing components into a real-time pipeline introduces practical challenges. Real-time systems must operate under

constraints involving inference latency, computational resources, memory consumption, and synchronization between independently processed signals. Sequential processing of audio preprocessing, speech representation, ASR, emotion classification, and linguistic analysis can increase end-to-end latency. Furthermore, streaming or short-segment processing can provide incomplete conversational context, while background noise, speaker variability, recording conditions, and recognition errors can propagate between processing stages. These factors create a practical trade-off between recognition quality, computational cost, and response latency in real-time audio systems.

To address these challenges, this work proposes a **Human State Representation Engine (HSRE)** as an intermediate structured reasoning layer between multimodal speech perception and downstream conversational systems. The proposed framework integrates evidence obtained from speech representation, automatic speech recognition, emotion detection, linguistic and intent analysis, and conversational context. These heterogeneous sources are normalized and fused before being processed through rule-based reasoning and probabilistic state scoring. The resulting representation describes multiple dimensions of the current conversational state, including **affective state, cognitive state, interaction state, and communication need**.

Unlike systems that directly map speech to a single emotion category, the proposed approach separates evidence extraction from state inference. This separation allows individual observations to retain their uncertainty while enabling multiple sources of evidence to contribute to the final state representation. The resulting Human State Representation (HSR) is produced as a structured JSON object, providing an interpretable intermediate representation that can subsequently be consumed by downstream conversational applications. The current implementation does not directly generate responses or employ a large language model; instead, it investigates whether structured multimodal state representation can provide a more informative and interpretable basis for real-time emotion-aware conversational systems.

## II. RELATED WORK

### *A. Self-Supervised Speech Representation Learning*

Self-supervised pretext tasks allow a model to learn general-purpose acoustic representations without transcribed labels. HuBERT formulates this as masked prediction of offline k-means cluster assignments over frame-level acoustic features, forcing the encoder to model both short-term acoustic structure and long-range temporal dependencies [1]. WavLM extends this masked-prediction objective with a denoising formulation that additionally exposes the model to overlapping speech and noise during pre-training, improving robustness for a broader "full-stack" set of speech-processing tasks beyond ASR, including speaker- and paralinguistic-oriented tasks [2]. These two families of self-supervised encoders motivate the choice of a HuBERT-based backbone for the speech-representation module of the proposed pipeline, since the resulting embeddings are expected to retain non-lexical information relevant to emotion in addition to phonetic content.

### *B. Speech Emotion Recognition with Self-Supervised Features*

Building directly on self-supervised encoders, Morais et al. evaluate an upstream-downstream configuration in which features pooled from a pretrained self-supervised model are passed to a lightweight downstream classifier for categorical emotion recognition, reporting that such features are competitive with, and often exceed, hand-crafted acoustic feature sets [3]. Pepino et al. similarly fine-tune and probe wav2vec 2.0 embeddings for emotion recognition, showing that contextual self-supervised representations transfer well to the emotion domain even though the pretext task itself is emotion-agnostic [4]. These results are consistent with, and precede, older convolutional-attention baselines such as Neumann and Vu's attentive CNN evaluated on IEMOCAP, which established that learned, attention-weighted acoustic representations outperform static hand-crafted descriptors for SER [9]. Collectively, this line of work supports the design decision, in the proposed system, to attach a lightweight fine-tuned classification head to a frozen or lightly fine-tuned HuBERT encoder rather than training an emotion model from raw waveforms.

### *C. Multimodal and Bimodal Emotion Fusion*

Atmaja et al. survey bimodal SER approaches that fuse acoustic and linguistic information, where the linguistic stream is typically obtained from an ASR transcript, and report that fusion consistently improves over acoustic-only systems, while also cataloguing the fusion strategies (feature-level, decision-level, and hybrid) used across the literature [5]. More recent systematic reviews broaden this picture to full multimodal emotion recognition combining audio, visual, and text modalities [6], and to multimodal emotion recognition in conversation specifically, surveying methods, benchmark datasets, and open challenges for dialogue-level affect modeling [7]. Hazarika et al.'s MISA framework operationalizes fusion by learning both modality-invariant and modality-specific subspaces before combining them for sentiment prediction, illustrating a concrete architecture for principled multimodal fusion [12]. The evidence-fusion stage of the proposed HSRE is informed by this literature: rather than fusing raw modality features, it fuses already-abstracted evidence (emotion probabilities, transcript-derived intent, and conversational context) at the decision/state level, which keeps the fusion step interpretable and inexpensive at inference time.

### *D. Conversational Context and Dialogue-Level State Modeling*

Because a single utterance is often ambiguous in isolation, several works model emotion as a function of the surrounding conversation. DialogueRNN tracks the states of individual speakers and the global conversational context using recurrent party- and emotion-state modules, demonstrating that dialogue-level context materially improves per-utterance emotion classification [13]. The Emotional Chatting Machine similarly conditions response generation on both content and an explicit emotion category, using internal and external memory to keep the expressed emotion consistent with context [14]. These works motivate the inclusion of a lightweight conversation-memory module in Phase 1 of the proposed system, which supplies recent transcripts, emotions, intents, and prior HSR states as contextual evidence to the HSRE, without requiring a full generative dialogue model.

### *E. Empathetic and Emotion-Aware Dialogue Systems*

Rashkin et al. introduce the EMPATHETICDIALOGUES benchmark and show that conditioning dialogue systems on the interlocutor's emotion improves perceived empathy of generated responses [10]. Liu et al. extend this direction with the ESConv dataset for emotional-support conversations, showing that effective support additionally requires selecting an appropriate support strategy for the current stage of the conversation, not just recognizing the emotion present [11]. Both works reinforce the premise of this paper: an emotion label alone under-specifies what a conversational system should do next, which motivates the HSRE's explicit "communication need" dimension (e.g., clarification, reassurance, empathy, concise vs. detailed response) as a direct, structured analogue of a support-strategy decision, produced deterministically rather than by a generative model.

### *F. Automatic Speech Recognition for Transcript Extraction*

Because linguistic evidence in the proposed pipeline is derived from an automatic transcript, transcription quality directly bounds the quality of downstream intent and semantic evidence. Whisper demonstrates that large-scale weak supervision on multilingual and multitask web-scraped audio-text pairs yields an ASR system that is robust to accent, background noise, and domain shift without task-specific fine-tuning [8]. This robustness motivates the use of Whisper Small as the transcription component of Phase 1, trading some accuracy relative to larger Whisper variants for reduced inference latency, which is a requirement for real-time operation.

### *G. Affective Computing Foundations*

Finally, the broader premise that computational systems can and should recognize, interpret, and respond appropriately to human affect traces back to Picard's foundational treatment of affective computing [15]. The present work operationalizes this premise for a real-time, speech-only setting, while deliberately constraining the output to a structured, non-diagnostic, explainable state representation rather than a free-form affective judgment.

## III. RESEARCH GAP AND MOTIVATION

Three observations from Section II motivate the design of the proposed system. First, most SER literature, including self-supervised approaches [1]–[4], [9], optimizes for a single categorical or dimensional emotion label per utterance; the richer conversational state that determines an appropriate system response — what the user needs, not merely how they feel — is typically left to a separate, often generative, downstream component. Second, multimodal and dialogue-context fusion studies [5]–[7], [12], [13] demonstrate that fusing evidence across modalities and turns improves recognition quality, but rarely expose the fused result as an explainable, structured intermediate representation that a non-expert or a downstream rule-based system can inspect and reason about directly. Third, emotion-aware and empathetic dialogue systems [10], [11], [14] increasingly rely on large generative language models to translate recognized affect into a response strategy, which introduces non-determinism, higher inference latency, and reduced auditability — properties that are undesirable for a real-time, safety-conscious human-state layer.

This paper addresses the gap between low-level perception (transcript, embeddings, emotion probabilities, intent) and high-level conversational application by proposing the Human State Representation Engine (HSRE): a deterministic, explainable, non-generative reasoning layer that fuses multi-source evidence using explicit rules and configurable probabilistic scoring to produce a structured, four-dimensional Human State Representation in real time.

## IV. PROPOSED SYSTEM

The proposed system is designed around five explicit goals derived from the motivation in Section III: (i) **modularity** — every model and reasoning component lives in its own folder/module and communicates through typed, structured objects, so any component can be replaced without touching the others; (ii) **explainability** — every inferred state is traceable to explicit rules and/or a transparent weighted score, never to an opaque generative step; (iii) **determinism** — identical evidence always yields identical reasoning output, which is a prerequisite for auditability; (iv) **real-time operability** — all models are loaded once at startup and inference is organized so that Phase 1 extraction and Phase 2 reasoning can be profiled and optimized independently; and (v) **no large language model at any stage**, so that the

system's behavior remains fully specified by its configuration and code rather than by an external generative model.

Fig. 1 summarizes the resulting architecture. The system is organized into two phases. Phase 1 (Section V) is purely extractive: it converts raw audio into a standardized evidence object containing a transcript, speech embeddings, an emotion probability distribution with confidence and intensity, and an intent/semantic estimate, augmented with short-term conversational history. Phase 1 performs no human-state reasoning. Phase 2, the Human State Representation Engine (Section VI), is purely reasoning: it consumes the Phase 1 evidence object (and only that object) and produces the final structured Human State Representation. This separation means the HSRE can be developed, tested, and reasoned about independently of which underlying speech, ASR, or embedding models are deployed upstream.

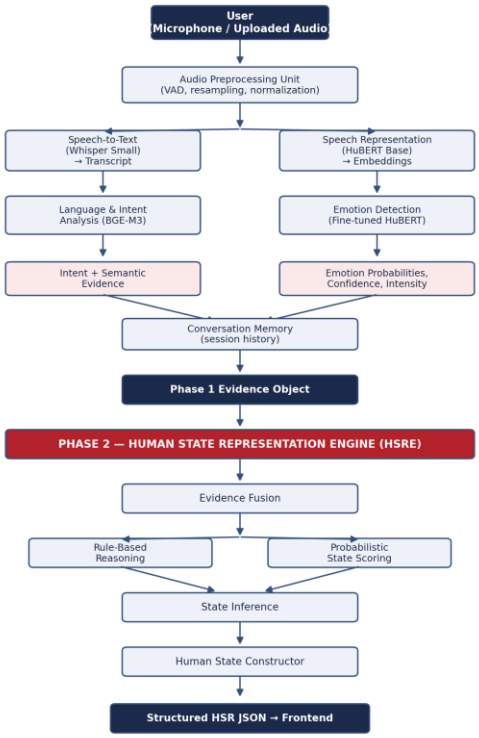



Fig. 1. End-to-end system architecture. Phase 1 (audio preprocessing through conversation memory) produces a standardized evidence object; Phase 2 (HSRE) fuses this evidence with rule-based and probabilistic reasoning to produce the structured Human State Representation.

The system is deployed as a two-part application: an Express.js frontend handles audio capture/upload and result rendering in the browser, while a FastAPI backend hosts all machine-learning inference (Phase 1) and the HSRE reasoning pipeline (Phase 2) behind a single analysis endpoint (Section VII).

## V. PHASE 1: AUDIO PROCESSING AND INFORMATION EXTRACTION

Phase 1 is responsible only for converting raw audio into structured, low-level evidence; it performs no human-state reasoning. It comprises six modules, each isolated in its own package: audio preprocessing, speech representation, speech-to-text, emotion detection, language and intent analysis, and conversation memory. Table I summarizes the model or method selected for each module.

**TABLE I**
*Module-to-Model Summary*

| Module | Selected Model / Method |
|---|---|
| Audio Preprocessing | Silero VAD + Librosa / torchaudio |
| Speech Representation | HuBERT Base [1] |
| Speech-to-Text | Whisper Small [8] |
| Emotion Detection | Fine-tuned HuBERT + classifier head [1],[3],[4] |
| Intent / Semantics | BGE-M3 embeddings + similarity/classification |
| Conversation Memory | Lightweight session-based memory |
| Human State Representation | Rule-based + probabilistic reasoning (HSRE) |
| Prompt Generation | Not implemented |
| LLM | Not implemented |

### *A. Audio Preprocessing*

Raw microphone or uploaded audio (WAV, MP3) is converted to a standardized form before any model consumes it: the signal is downmixed to mono, resampled to 16 kHz, and amplitude-normalized. Silero VAD is then applied to detect speech regions and trim non-speech silence, after which optional noise filtering and frame/windowing are applied. The module outputs a standardized processed_audio.wav together with metadata (sample rate, channel count, total duration, and speech duration), which downstream modules use to allocate compute proportionally to the amount of actual speech present.

### *B. Speech Representation*

A HuBERT Base encoder [1] extracts contextual speech embeddings from the processed audio via a single function-level interface, extract_speech_embedding(audio), so that the specific encoder implementation remains fully encapsulated and swappable (e.g., for WavLM [2]) without affecting any other module. These embeddings feed the emotion-detection module and are not exposed directly to the frontend.

### *C. Speech-to-Text*

Whisper Small [8] transcribes the processed audio into text. Whisper is used exclusively for automatic speech

recognition; consistent with the no-LLM constraint of this work, it is never used for downstream reasoning, summarization, or generation. Whisper Small is selected over larger Whisper variants specifically to bound transcription latency for real-time operation, at the cost of some transcription accuracy that is quantified in Section IX.

### *D. Emotion Detection*

A lightweight classification head fine-tuned on top of HuBERT representations [1], following the upstream-downstream methodology validated for SER by Morais et al. [3] and Pepino et al. [4] and consistent with attention-based acoustic classifiers such as [9], predicts an emotion label together with (i) a full probability distribution over the trained emotion classes, (ii) a scalar confidence, and (iii) a scalar intensity. Critically, the module never collapses this output to a single label before it reaches the HSRE: the full probability distribution is preserved because Phase 2 treats it as weighted evidence rather than a ground-truth judgment. If the trained classifier's label set does not include a higher-level state such as "frustrated," that state is instead derived downstream by the HSRE from the combination of base emotion probabilities, transcript content, and conversational context (Section VI).

### *E. Language and Intent Analysis*

BGE-M3 embeddings are used, strictly as an embedding model rather than a generative one, to obtain a semantic representation of the transcript and to perform intent classification by similarity against a fixed, predefined set of intent descriptions (e.g., asking_information, requesting_help, complaint, confirmation, disagreement, feedback, greeting, farewell, troubleshooting, unknown). This keeps intent detection explainable — each prediction can be traced to a nearest-neighbor intent description and a similarity score — and avoids any generative language modeling step.

### *F. Conversation Memory*

A lightweight, session-scoped, in-memory store retains the information needed for the current conversation only: recent transcripts, emotions, intents, HSR states, and interaction outcomes. No vector database or persistent store is introduced in this implementation, consistent with the goal of a simple, working research prototype rather than production infrastructure; this can be revisited if conversation length or retrieval requirements grow (Section XII).

The outputs of modules A–F are merged into one standardized Phase 1 evidence object, illustrated below, which becomes the sole input to Phase 2.

```
{
  "audio": { "sample_rate": 16000, "duration":
4.82 },
  "transcript": "I have tried this several times
                 but it is still not working.",
  "emotion": {
    "label": "angry", "confidence": 0.91,
    "intensity": 0.84,
    "probabilities": { "neutral": 0.04, "happy":
0.02,
      "sad": 0.08, "angry": 0.61, "fear": 0.05,
      "surprise": 0.02, "disgust": 0.18 }
  },
  "intent": { "label": "requesting_help",
"confidence": 0.87 },
  "speech_representation": { "embedding_available":
true }
}
```

## VI. PHASE 2: THE HUMAN STATE REPRESENTATION ENGINE (HSRE)

The HSRE (Fig. 2) is the primary contribution of this work. It is implemented as a completely separate module from the emotion model, the ASR model, and the frontend, and it accepts only the standardized Phase 1 evidence object — never raw audio — which keeps Phase 2 independently testable and independently replaceable. The HSRE performs no LLM inference at any stage; every inferred state is produced by explicit rules, configurable weighted scoring, or a combination of the two.

### *A. 6.1 Evidence Fusion*

The evidence-fusion stage normalizes and aligns the Phase 1 evidence (emotion probabilities, confidence, and intensity; the transcript and its intent/semantic evidence; and contextual evidence from conversation memory), assigning each source a confidence-adjusted weight and packaging the result into a unified evidence object with four named evidence groups — emotion_evidence, linguistic_evidence, intent_evidence, and contextual_evidence — consumed by the reasoning layer.

### *B. 6.2 Reasoning Layer*

Human-state inference is performed by two complementary, explainable mechanisms.

**1) Rule-Based Reasoning:** Rules are explicit IF–THEN statements over the fused evidence, for example:

```
IF emotion = angry AND intent = requesting_help
THEN communication_need = clarification

IF negative_emotion_confidence > 0.75 AND
   previous_interaction contains repeated_failure
THEN interaction_state = frustrated_interaction

IF emotion_confidence < threshold
THEN reduce influence of emotion evidence
```

**2) Probabilistic State Scoring:** No single prediction is treated as absolute. Each candidate state receives a weighted score computed from the fused evidence, e.g., for frustration:

```
frustration_score =
    emotion_score      × emotion_weight
  + intent_score       × intent_weight
  + contextual_score   × context_weight
```

All weights (e.g., emotion_weight = 0.45, intent_weight = 0.25, context_weight = 0.20, linguistic_weight = 0.10 in the reference configuration) are stored in a single configuration file rather than hard-coded across the codebase, so that the evidence-to-state mapping can be retuned without modifying reasoning logic. The reasoning layer's output for each candidate state reports the inferred state, its numeric score, a confidence value, and the specific supporting evidence that produced it — the basis for the system's explainability.

### *C. 6.3 State Inference*

The reasoning layer infers states along four complementary dimensions, summarized in Table II. These are deliberately kept at the level of conversational/affective estimation: the system must never claim to have diagnosed depression, anxiety disorder, PTSD, or any other psychiatric condition — it estimates conversational and affective state from available acoustic, linguistic, and contextual signals only, a constraint enforced throughout the reasoning and state-construction code.

**TABLE II**

*HSRE State Dimensions and Possible Inferred States*

| Dimension | Possible Inferred States |
|---|---|
| Affective State | calm, happy, sad, angry, fearful, frustrated, excited, neutral |
| Cognitive State | confused, focused, uncertain, confident, information_seeking |
| Interaction State | casual, help_seeking, repeated_failure, disagreement, escalation, normal_interaction |
| Communication Need | reassurance, clarification, concise_response, detailed_response, empathy, encouragement, direct_answer |

### *D. 6.4 Human State Constructor*

The Human State Constructor assembles the outputs of the reasoning layer into one standardized HSR object with four state blocks plus a reasoning trace listing the supporting evidence, illustrated below for the running example.

```
{
  "affective_state": { "emotion": "angry",
    "confidence": 0.91, "intensity": 0.84,
    "frustration_score": 0.82 },
  "cognitive_state": { "intent": "requesting_help",
    "confidence": 0.87, "uncertainty": 0.32 },
  "interaction_state": { "state":
"repeated_failure",
    "confidence": 0.79 },
  "communication_need": { "primary":
"clarification",
    "secondary": "empathy",
    "recommended_detail": "step_by_step" },
  "reasoning": { "supporting_evidence": [
      "high negative emotional intensity",
      "help-seeking intent",
      "repeated unsuccessful attempts" ] }
}
```

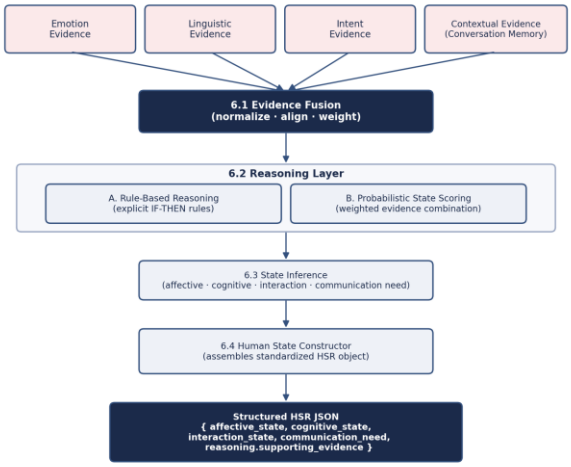



Fig. 2. Internal structure of the Human State Representation Engine (HSRE): evidence fusion, the dual rule-based / probabilistic reasoning layer, state inference, and the human state constructor.

## VII. SYSTEM IMPLEMENTATION AND ARCHITECTURE

The system is realized as a two-part monorepo application. The frontend is an Express.js web application responsible for microphone recording, audio upload, and rendering the structured result. The backend is a FastAPI service that hosts every machine-learning model and the HSRE, organized so that models are loaded once at application startup rather than per request, and every module is kept independent (audio_preprocessing/, speech_representation/, speech_to_text/, emotion_detection/, language_analysis/, conversation_memory/, and the fully separate hsre/ package with its own evidence_fusion/, reasoning/, state_constructor/, and schemas/ sub-packages).

The backend exposes a single primary analysis endpoint plus a health check:

```
POST /api/analyze
  in:  audio file
  out: audio metadata, transcript,
       emotion analysis, intent analysis,
       human state representation (HSR)

GET  /api/health   ->  { "status": "healthy" }
```

A representative response from *POST /api/analyze* is:

```
{
```

```
  "status": "success",
  "input": { "source": "microphone", "duration":
5.2 },
  "transcript": "I have tried this several times
                 but it is still not working.",
  "emotion_analysis": { "emotion": "angry",
    "confidence": 0.91, "intensity": 0.84 },
  "intent_analysis": { "intent": "requesting_help",
    "confidence": 0.87 },
  "human_state": {
    "affective_state": { "emotion": "angry",
      "frustration_score": 0.82 },
    "cognitive_state": { "intent":
"requesting_help" },
    "interaction_state": { "state":
"repeated_failure" },
    "communication_need": { "primary":
"clarification",
      "secondary": "empathy",
      "recommended_detail": "step_by_step" }
  }
}
```


The implementation follows a fixed set of engineering requirements intended to keep the prototype modular, inspectable, and free of unnecessary infrastructure:

- Every module is independent; model logic is never placed directly in main.py.
- HSRE logic is never mixed with emotion-detection logic.
- The HSRE accepts only structured evidence, never raw audio.
- All models are loaded once at startup, not per request.
- Model paths and HSRE weights live in configuration files, not inline code.
- API inputs and outputs use typed schemas.
- Invalid or unsupported audio is handled with explicit error responses.
- User audio is not stored permanently unless explicitly required.
- No LLM, vector database, Docker, authentication layer, or cloud deployment is introduced unless specifically required beyond this prototype.

## VIII. EXPERIMENTAL SETUP

This section specifies the protocol under which the emotion-detection classifier, the intent classifier, and the end-to-end HSRE pipeline are evaluated. The exact dataset(s), splits, and hardware used for the reported experiments are project-specific choices left to be finalized during evaluation and are marked below.

### *A. Dataset*

Emotion-detection fine-tuning and evaluation require a labeled speech-emotion corpus (e.g., a public benchmark such as IEMOCAP, RAVDESS, CREMA-D, or a project-collected dataset). Intent-classification evaluation additionally requires transcript-level intent annotations for the chosen intent taxonomy (Section V-E).

**TABLE III**
*Dataset Statistics*

| Split | # Utterances | # Speakers | Emotion Classes |
|---|---|---|---|
| | | | |
| | | | |
| | | | |

### *B. Emotion Classifier Training Configuration*

The emotion-detection classification head (Section V-D) is fine-tuned on top of frozen or lightly fine-tuned HuBERT Base representations. Table IV records the exact training configuration once finalized.

**TABLE IV**
*Emotion Classifier Training Hyperparameters*

| Hyperparameter | Value |
|---|---|
| Base encoder | HuBERT Base [1] |
| Classification head | |
| Optimizer | |
| Learning rate | |
| Batch size | |
| Epochs / steps | |
| Train / val / test split | |

### *C. Evaluation Metrics*

Emotion recognition is evaluated using weighted accuracy (WA), unweighted accuracy (UA), and macro-averaged F1, consistent with standard SER reporting practice. Intent classification is evaluated using classification accuracy and macro-F1 over the intent taxonomy of Section V-E. HSRE state-inference quality is evaluated qualitatively via case studies (Section IX-C) and, where ground-truth conversational-state labels are available, quantitatively via agreement with human-annotated reference states. End-to-end real-time performance is evaluated via per-module and total pipeline latency (Section IX-D).

### *D. Hardware and Software Environment*

⚙ **TO BE FILLED BY AUTHOR:** Specify the hardware used for training and inference (CPU/GPU model, RAM), software versions (Python, PyTorch/HuggingFace Transformers, FastAPI, Node.js/Express versions), and whether inference was measured on CPU or GPU.

## IX. RESULTS AND DISCUSSION

This section reports the empirical evaluation of the pipeline described in Sections V–VII, following the protocol of Section VIII. All numeric results below are placeholders to be populated once experiments are run on the finalized dataset and hardware.

### *A. Emotion Recognition Performance*

Table V compares the fine-tuned HuBERT emotion classifier used in Phase 1 (Section V-D) against baseline SER approaches drawn from the related-work literature (e.g., wav2vec 2.0 embeddings [4], attentive CNN [9]).

**TABLE V**
*Emotion Recognition Performance*

| Model | WA (%) | UA (%) | Macro-F1 |
|---|---|---|---|
| Fine-tuned HuBERT (proposed) | | | |
| Baseline (e.g., wav2vec 2.0 [4]) | | | |
| Baseline (e.g., attentive CNN [9]) | | | |

**▦ IMAGE PLACEHOLDER**
Confusion matrix of predicted vs. ground-truth emotion classes on the held-out test set — insert the normalized confusion-matrix heatmap generated from the trained emotion classifier's test predictions.

Fig. 3. Confusion matrix of predicted vs. ground-truth emotion classes on the held-out test set

### *B. Intent Classification Performance*

Table VI reports per-class precision, recall, and F1 for the BGE-M3 similarity-based intent classifier of Section V-E against the annotated intent taxonomy.

**TABLE VI**
*Intent Classification Performance*

| Intent Class | Precision | Recall | F1 |
|---|---|---|---|
| asking_information | | | |
| requesting_help | | | |
| complaint | | | |
| confirmation | | | |
| … (remaining classes) | | | |
| Macro Average | | | |

### *C. HSRE State-Inference Case Studies*

Table VII illustrates qualitative HSRE behavior on representative inputs, pairing the Phase 1 evidence (emotion, intent) with the resulting communication-need recommendation from Phase 2. The first row reproduces the worked example used throughout Sections V–VII; remaining rows are to be populated with additional recorded/transcribed examples spanning each interaction-state category of Table II.

**TABLE VII**
*Representative HSRE Case Studies*

| Transcript (excerpt) | Emotion | Intent | Inferred Communication Need |
|---|---|---|---|
| I have tried this several times but it is still not working. | angry | requesting_help | clarification + empathy |
| | | | |
| | | | |

**⚙ TO BE FILLED BY AUTHOR:** Add 6–10 additional case-study rows covering each interaction_state and communication_need category from Table II (casual, help_seeking, repeated_failure, disagreement, escalation, normal_interaction), and discuss any cases where the rule-based layer and the probabilistic scorer disagreed.

**▦ IMAGE PLACEHOLDER**
Screenshot of the Express.js frontend showing microphone/upload controls and the rendered Human State Representation (affective, cognitive, interaction state, and communication need) for a sample utterance.

Fig. 4. Screenshot of the Express.js frontend showing microphone/upload controls and the rendered Human State Representation (affective, cognitive, interaction state, and communication need) for a sample utterance.

### *D. Latency and Real-Time Performance*

Table VIII and Fig. 4 report mean per-module and total end-to-end latency, measured from receipt of a finalized audio clip at the FastAPI endpoint to the returned HSR JSON, averaged over the test set of Section VIII-A.

**TABLE VIII**
*Per-Module and End-to-End Latency*

| Module | Mean Latency (ms) | % of Total |
|---|---|---|
| Audio Preprocessing | | |

| Module | Mean Latency (ms) | % of Total |
|---|---|---|
| Speech Representation (HuBERT) | | |
| Speech-to-Text (Whisper Small) | | |
| Emotion Detection | | |
| Language & Intent Analysis (BGE-M3) | | |
| HSRE (fusion + reasoning + construction) | | |
| Total end-to-end | | |

**▦ IMAGE PLACEHOLDER**

Bar chart of mean per-module latency (ms) for audio preprocessing, HuBERT speech representation, Whisper Small transcription, emotion detection, BGE-M3 intent analysis, and HSRE reasoning, alongside total end-to-end latency — insert the measured latency bar chart with error bars.

Fig. 5. Bar chart of mean per-module latency (ms) for audio preprocessing, HuBERT speech representation, Whisper Small transcription, emotion detection, BGE-M3 intent analysis, and HSRE reasoning, alongside total end-to-end latency

### *E. Ablation: Rule-Based vs. Probabilistic Contributions*

To isolate the contribution of each reasoning mechanism in Section VI-B, the HSRE is evaluated in three configurations: rule-based reasoning only, probabilistic scoring only, and the full combined pipeline.

**⚙ TO BE FILLED BY AUTHOR:** Report state-inference agreement (with human-annotated reference states, if available) for each of the three HSRE configurations, and discuss which conversational-state dimensions (affective, cognitive, interaction, communication need) benefit most from combining rules with probabilistic scoring.

## X. LIMITATIONS

The proposed system has several limitations that constrain the interpretation of its results. First, the emotion classifier operates over a fixed, trained label set; higher-order states such as "frustration" are derived by the HSRE rather than predicted directly, so their quality is bounded by the quality and coverage of the hand-authored rule set and the chosen scoring weights (Section VI-B), which currently require manual curation and domain expertise to extend to new use cases or languages. Second, the intent taxonomy of Section V-E is a fixed, predefined set of ten categories; utterances whose intent falls outside this set are mapped to an "unknown" category rather than being modeled directly. Third, the reported architecture has been evaluated primarily on single-speaker, single-turn or short-session interactions. Fourth, as with any speech pipeline, performance is sensitive to microphone quality, background noise, accent, and speaking rate, each of which can degrade ASR transcript quality and therefore the linguistic and intent evidence available to the HSRE.

**⚙ TO BE FILLED BY AUTHOR:** State how multi-speaker and long-session conversation-memory behavior was evaluated (if at all), and summarize accent/noise/microphone robustness findings, or note these as directions for future testing if not yet measured.

Finally, the HSRE's outputs are estimates of conversational and affective state derived from surface acoustic, linguistic, and contextual signals; they are not, and are not intended to be, clinical or diagnostic assessments of any kind.

## XI. ETHICAL CONSIDERATIONS

Several design choices in this work are deliberately motivated by ethical and safety considerations rather than by performance alone:

- No clinical claims: the system never labels a psychiatric condition (e.g., depression, anxiety disorder, PTSD); Section VI-C enforces this constraint at the state-inference level.
- Data minimization: user audio is not stored permanently unless explicitly required by a deployment, and conversation memory is scoped to the active session (Section V-F).
- Transparency and explainability: every inferred state is traceable to explicit rules and/or a disclosed weighted score and its supporting evidence (Section VI-B–D), rather than to an opaque generative process.
- Consent and disclosure: deployments of this system should inform users that their speech is being analyzed for emotional and conversational state, and should obtain appropriate consent before recording.

**⚙ TO BE FILLED BY AUTHOR:** Report any fairness/bias evaluation performed across speaker demographics (accent, gender, age) present in the evaluation dataset, and note any observed disparities in emotion- or intent-recognition accuracy across these groups.

## XII. CONCLUSION AND FUTURE WORK

This paper presented a real-time, modular, and fully explainable pipeline that converts raw speech into a structured Human State Representation without invoking a large language model at any stage. Phase 1 extracts transcript, speech-representation, emotion, and intent evidence using HuBERT-based encoders [1], Whisper Small [8], and BGE-M3 embeddings, while Phase 2, the Human State Representation Engine, fuses this evidence through explicit

rule-based reasoning and configurable probabilistic scoring to infer four complementary state dimensions: affective state, cognitive state, interaction state, and communication need. This design directly addresses the gap identified in Section III between single-label emotion recognition and the richer, explainable conversational state required by downstream emotion-aware applications [10], [11], [13], [14], while avoiding the latency and auditability costs of a generative reasoning layer.

Future work includes: (i) extending evidence fusion to visual/multimodal cues in line with recent multimodal emotion-recognition surveys [6], [7]; (ii) validating the HSRE's rule set and scoring weights across larger and more diverse, multi-speaker, multilingual datasets; (iii) exploring iterative or learned refinement of the rule/weight configuration itself while preserving explainability; (iv) on-device or edge optimization of the Phase 1 encoders to further reduce end-to-end latency; and (v), as noted in the original design constraints, optionally passing the structured HSR — rather than raw audio or transcript — as context to an external LLM in a future system, keeping the perception-and-reasoning layer described in this paper fully deterministic and auditable regardless of how any downstream generative component is used.

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